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$$\begin{aligned} & \lim_{x \rightarrow 0} \left(\frac{\sqrt{2+x} - \sqrt{2-x}}{x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(\sqrt{2+x} - \sqrt{2-x})(\sqrt{2+x} + \sqrt{2-x})}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{2+x}^2 - \sqrt{2-x}^2}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(2+x) - (2-x)}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2x}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2}{\sqrt{2+x} + \sqrt{2-x}} \right) \\ &= \frac{2}{\sqrt{2+0} + \sqrt{2-0}} \\ &= \frac{2}{\sqrt{2} + \sqrt{2}} \\ &= \frac{2}{2\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

References